

Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

Reasoning + Answer:

<reasoning process here> CD 2 1 is a receptor on B cells that plays a critical role in B cell activation and antigen recognition . It is part of the complement receptor complex and is involved in enhancing B cell responses to antigens by binding to complement components (like C 3 d) that are associated with pathogens . This interaction helps B cells detect and respond to viruses and other pathogens more effectively .

One key path ogen that heavily relies on the CD 2 1 -mediated signaling pathway for efficient infection and immune evasion is the Epstein -B arr virus (EB V). EB V has been shown to exploit the B cell environment , and the interaction between EB V and the complement system , in conjunction with CD 2 1 , plays a significant role in establishing and maintaining latency in B cells .

Not ably , EB V enters B cells and relies on B cell function for replication . By knocking out CD 2 1 , the B cells may be impaired in their ability to recognize EB V -infected cells or respond to complement -co ated viral antig ens , thereby disrupting viral replication or establishment .

Me as les , HIV , and HPV do not rely primarily on CD 2 1 signaling or complement receptor pathways in their pathogenesis . HIV targets CD 4 + T cells and macro ph ages , while HPV primarily infects epith elial cells . Me as les virus infects epith elial and dend ritic cells , and its pathogenesis is not heavily dependent on CD 2 1 -mediated B cell responses .

Thus , the genetic modification of B cells lacking CD 2 1 would most significantly impact the pathogenesis of Epstein -B arr virus (EB V). </reasoning process> < answer > A </answer>

